

NOMBALI MAFA

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SUMMARY

I am a full stack developer and machine learning practitioner with a strong track record in building impactful technology solutions. I work at the intersection of software engineering and AI, with a focus on health technology, conservation, and social impact. I led student mentorship at the TechXplo Foundation and serve on the Hackathon planning committee for the Deep Learning IndabaX South Africa.

EXPERIENCE

- 1st Place: W3Node Node Engine Hackathon 2026, Gaming & Entertainment Track ("Spaza Tycoon")
- 1st Place: GirlCode Hackathon 2025, National Winner (Johannesburg & South Africa)
- 1st Place: TPC/CBE University of Johannesburg Hackathon (2023)
- 1st Place: Campus Innov8 Hacker/Makerthon (2023)
- 1st Place: UJ CBE/TPC Hackathon (2023)
- 1st Runner Up: GirlCode Hackathon (2023)
- 1st Runner Up: EPF EmpowerHer Hackathon (2023)
- 2nd Runner Up: EPF x NCIS Hackathon (2023).

Hackathon Organiser

Deep Learning IndabaX South Africa

April 2026 - Present, South Africa

- Spearheaded the strategic planning and execution of two hackathon tracks (Beginner and Advanced) for the Deep Learning IndabaX South Africa 2026 conference.
- Designed engaging, practical Machine Learning workshops tailored to varying experience levels, driving active participation for a conference of **200+** attendees.
- Played a key role in advancing the organisation's mission to spread ML capacity across South Africa by creating accessible, community-driven technical competitions that fostered skill development and networking.

Full Stack Developer & Student Mentorship Lead

TechXplo

July 2024 - July 2026, Randburg, South Africa

- Architected and deployed scalable full-stack web and mobile applications using React, Node.js, and Supabase, successfully delivering solutions for 4 commercial and 2 non-profit clients.
- Led the end-to-end development of *InsureLog*, a specialised insurance management training platform used to upskill students and employees on personal and commercial lines.
- Directed the TechXplo Foundation mentorship program, successfully guiding **15** students through real-world industry projects from conceptualisation to production-ready deployment.
- Streamlined project delivery by collaborating directly with stakeholders to define requirements and optimise sprint planning, reducing average feature deployment time by 95%.

Academic Tutor: Development Software 1A and 1B & Development Software 2A and 2B

University of Johannesburg

February 2025 - November 2025, Johannesburg, South Africa

- Taught first-year students Python and C# programming fundamentals to beginner level OOP principles.
- Taught second-year students HTML, CSS, JavaScript, PHP, MySQL for traditional website development, ReactJS for Web App development, and mobile app development using React Native and Firebase as backend.
- Facilitated hands-on programming tutorials for 1st- and 2nd-year university students, bridging the gap between theoretical concepts and practical application.
- Delivered comprehensive instruction across multiple tech stacks, guiding first years programming basics through to OOP fundamentals (Python, C#) and advancing 2nd-year students through full-stack web and mobile development (HTML/CSS, JS, ReactJS, PHP, MySQL, React Native, Firebase).
- Improved student comprehension of coding theory and complex coding architecture.

Student Peer Mentor

University of Johannesburg

February 2024 - November 2025, Johannesburg, South Africa

- Mentored a dedicated group of **10** first-year students in 2024 and 7 students in 2025, providing personalized guidance and resources to ensure a seamless transition into university academia.
 - Conducted weekly one-on-one advisory sessions to track progress and execute a structured mentorship curriculum, maintaining a 90% mentee retention rate.
 - Equipped students with essential academic tools, study strategies, and campus resources, resulting in high academic achievement and consistent positive feedback from the mentee cohort.
 - Cultivated a supportive environment that fostered leadership, communication, and adaptability skills for incoming students.
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PROJECT

Community & Leadership

- Speaker: GirlCode 2026 International Girls in ICT Day Virtual Summit
- Speaker: UJ Methods Lab
- H.E.R DAO South Africa Representative
- SheFi Scholar: Season 16 (Crypto, DeFi & AI programme)
- Hackathon Planning Committee Member: Deep Learning IndabaX South Africa 2026.

Hackathon Wins

May 2023 – January 2026

- 1st Place: W3Node Node Engine Hackathon 2026, Gaming & Entertainment Track ("Spaza Tycoon")
- 1st Place: GirlCode Hackathon 2025, National Winner (Johannesburg & South Africa)
- 1st Place: TPC/CBE University of Johannesburg Hackathon (2023)
- 1st Place: Campus Innov8 Hacker/Makerthon (2023)
- 1st Place: UJ CBE/TPC Hackathon (2023)
- 1st Runner Up: GirlCode Hackathon (2023)
- 1st Runner Up: EPF EmpowerHer Hackathon (2023)
- 2nd Runner Up: EPF x NCIS Hackathon (2023).
- Secured first-place finishes in competitive hackathons, including the W3Node Node Engine Hackathon and the GirlCode National competition, demonstrating advanced problem-solving and rapid development skills.

EDUCATION

Diploma in Business Information Technology

University of Johannesburg • Johannesburg, South Africa • 2026

- Awarded funding for second and final year studies due to good grades.

CERTIFICATIONS

UJ Peer Mentorship Award

University of Johannesburg • 2025

- This certificate recognizes my hands-on experience guiding first-year students through their critical transition into university life. By providing personalized support and connecting mentees with essential resources, I developed strong empathy, patience, and active listening skills. Mentoring beginners and helping them build a solid foundation directly relates to my professional work, as it demonstrates my ability to effectively onboard new team members, explain foundational concepts clearly, and foster a supportive, collaborative environment.

Drone Detection Workshop

Opti-Num • 2025

- This workshop expanded my technical understanding of real-time monitoring and threat identification systems. Exploring drone detection technologies deepened my knowledge at the intersection of security and artificial intelligence. It provided practical insights into processing complex sensor data and understanding object detection mechanisms, which strengthens my ability to conceptualise secure, data-driven software architectures and broadens my general expertise in machine learning applications.

UJ Peer Mentorship Award

University of Johannesburg • 2024

- This certificate recognizes my hands-on experience guiding first-year students through their critical transition into university life. By providing personalized support and connecting mentees with essential resources, I developed strong empathy, patience, and active listening skills. Mentoring beginners and helping them build a solid foundation directly relates to my professional work, as it demonstrates my ability to effectively onboard new team members, explain foundational concepts clearly, and foster a supportive, collaborative environment.

Women in AI Summit

Google Developers Group • 2024

- This certificate highlights my proactive engagement with the artificial intelligence landscape and my commitment to fostering inclusive tech communities. Participating in this summit reinforced my knowledge of current AI trends and ethical considerations, which directly supports my work structuring machine learning hackathon tracks and guiding aspiring developers through industry projects. It demonstrates a dedication to continuous learning and the ability to integrate forward-thinking, diverse perspectives into software development and technical mentorship initiatives.

Youth Leadership and Climate Change

The Earthshot Prize • 2024

- This certificate underscores my commitment to applying technology toward environmental and social impact. My technical background in AI and software development regularly intersects with sustainability, such as when designing a machine learning framework for South African wildlife protection. Gaining this training through The Earthshot Prize enhances my ability to understand complex ecological challenges and lead purpose-driven tech projects that focus on climate resilience and sustainable innovation.

IBM Z Xplore: Fundamentals and Concepts

IBM • 2024

- This certificate provided me with foundational knowledge of enterprise computing, data management, and mainframe architecture, which are the backbone of the global banking and financial sectors. Given my strong interest in fintech and security, understanding IBM Z systems gives me a unique perspective on how large-scale, highly secure transactions are processed. It broadens my technical versatility beyond modern web and mobile development, equipping me with the architectural context needed to build robust, secure applications that can integrate with enterprise-level infrastructure.

Introduction to Data Science

Cisco • 2023

- This certificate establishes a foundational understanding of data manipulation, analysis, and visualization, which is essential for my ongoing work in machine learning and artificial intelligence. As a full-stack developer working on AI-driven applications and organizing machine learning initiatives, this knowledge enables me to effectively manage data pipelines, evaluate data-driven models, and seamlessly integrate complex datasets into the software solutions and platforms I build.

Enterprise Design Thinking Practitioner

IBM • 2023

- This certificate demonstrates my ability to apply a user-centric approach to software development. As a full-stack developer, understanding design thinking allows me to go beyond just writing code and focus on the actual needs of the end-user. It highlights my capability to collaborate with stakeholders, define core problems, empathise with users, and rapidly prototype solutions, ensuring that the web and mobile applications I build are both functional, accessible, and intuitive.

Great Learning Front End Development- HTML

Great Learning • 2023

- This certificate established my foundational understanding of web development by focusing on semantic document structure, accessibility, and form inputs. Even though I now work extensively with modern JavaScript frameworks like React and React Native, a solid grasp of core HTML is critical for building well-structured, accessible, and SEO-friendly components. It serves as the essential bedrock for the complex, responsive web applications I develop today.

SKILLS

Frontend: React, React Native, HTML, CSS, JavaScript

Backend: NodeJS, PHP, C#, Python, ExpressJS

Databases: Firebase, MySQL, SQL, Supabase

Data & ML: Python, Pandas, Scikit-learn, Random Forest, Linear Regression, Data Visualisation

Other: API Integration, Git/GitHub, Full Stack Web & Mobile App Development

REFERENCES

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